

CodonContext: a fail-closed CDS constraint tool with no expression claim

Candidate-specific working report, September 25, 2026. This project is separate from the aggregate item-27 manuscript. A later same-protein, one-host mRFP expression analysis now supplies real paired construct measurements, but does not test the original cross-host CAI-gain/yield question or establish a new design method. The useful software deliverable is a tested CDS-constraint and provenance checker that refuses to turn codon metrics into yield predictions.

Abstract

Codon adaptation index (CAI) is a descriptive measure of how supplied synonymous codon weights match a particular usage table. Raising a CAI value does not, by arithmetic alone, prove that a protein is expressed, secreted or functional. The original candidate proposed testing whether CAI gain predicts experimentally measured expression across hosts under held-out GC and secretion constraints. The original examples lacked paired construct measurements, and a later mRFP cohort supplies only one protein in one host; four source-header holdouts show unstable calibration (macro R2 -0.305). We therefore keep the fallback: a standard-code coding-sequence checker for protein identity, whole and sliding-window GC, forbidden DNA motifs, homopolymer runs and supplied host-specific relative codon weights. It refuses invalid inputs and always leaves `expression_prediction` null. Three exact UniProtKB/NCBI RefSeq protein-CDS pairs were retrieved and crosschecked; their CDS translations match the corresponding protein sequences. These real sequences test parsing and translation, not expression. The software gate passes; the proposed host-/protein-held-out CAI-gain biological question remains unassessable. The mRFP early-codon result reproduces a published theme and is not a new discovery.

1. Initial biological question and pivot

A codon optimization study would need multiple constructs encoding the same protein, measured protein yield in multiple expression hosts, and a defined separation of train and test by host, GC stratum, secretion pathway or laboratory. It would compare a CAI-only expectation to the observed yield after controlling for composition and context. A collection of coding-sequence accessions alone does not supply such a label. Codon choice also affects mRNA structure, ribosome dynamics, motif creation, regulatory elements and host stress, none of which is captured by a single CAI value.

Because that data gate was not met, the computational task is narrower. A user can submit a candidate and optional reference DNA CDS. The tool checks whether it encodes the same polypeptide and satisfies explicit, user-supplied sequence constraints. A lack of protein identity is a hard error. Constraint violations are returned as positions and values, not turned into a predicted expression rank. A user may supply relative codon adaptiveness weights for descriptive CAI arithmetic, but the tool will not invent a host table or measured response.

2. Algorithm and formulas

For a CDS of length L nucleotides, valid input requires $L > 0$, L divisible by 3, only A/C/G/T, initial ATG, a terminal stop codon TAA/TGA/TAG and no internal in-frame stop. Translation uses the standard nuclear/bacterial codon map. A candidate and reference must have identical translated amino-acid strings; the stop is omitted from the protein sequence hash. This is a sequence-preservation check, not a statement that the protein folds or behaves identically in different hosts.

Whole-sequence GC fraction is $(\text{count}(G) + \text{count}(C)) / L$. The optional sliding window of W nucleotides visits every position from 0 through $L - W$, inclusive, and compares each window's GC fraction to supplied minimum and maximum. A forbidden motif is matched against exact uppercase DNA text and its first zero-based position is reported. Homopolymer check flags a run longer than the supplied maximum. Limits must be finite and in $[0, 1]$ for GC, W must be an integer from 3 through L , and contradictory min/max ranges are rejected. Boolean and NaN values are refused; silently interpreting them as numbers could make a wrong construct appear feasible.

For each non-Met, nonsingleton synonymous codon c , the optional CAI is the exponential of the arithmetic mean of log relative-adaptiveness weight $w(c)$, where each required weight is in $(0, 1]$. The tool returns null when there are no variable synonymous codons. CAI gain is candidate CAI minus reference CAI only when both sequences and all necessary weights are supplied. This is an arithmetic difference, not a percentage expression gain. Supplying weights without their host and experimental origin can make a number look precise while remaining biologically uninterpretable.

3. Real accession provenance

`accession_audit.py` fetched exact UniProtKB records and the linked NCBI RefSeq nucleotide records. For each protein, it selected the GenBank CDS feature with the matching protein ID, extracted the DNA at annotated coordinates, translated it and required exact equality to the separately retrieved UniProtKB protein sequence. This includes reverse-complement feature handling where necessary. The results and SHA-256 response digests are in `accessions.json`.

UniProtKB protein	RefSeq protein	Nucleotide record	Protein length	CDS/UniProt match	Yield measured here?
P00722, E. coli beta-galactosidase	NP_414878.1	NC_000913.3	1,024 amino acids	Yes	No
P07102, E. coli AppA phytase	NP_415500.1	NC_000913.3	432 amino acids	Yes	No
P61769, human beta-2-microglobulin	NP_004039.1	NM_004048.4	119 amino acids	Yes	No

P07102 occurs in the earlier PhosGene project. Its re-use as a sequence-validity test is not a new independent biological discovery for either project. These three crosschecked pairs count as three individually retrieved pairs within this tool; they do not satisfy a 120-independent-accession program gate. NCBI RefSeq and UniProtKB are two distinct biological sources; Biopython parses GenBank, but is a local library rather than a third biological service.

4. Tool behavior and examples

The committed example.json contains synthetic short synonymous CDS inputs, a small illustrative adaptiveness table and constraint values. Run `python3 projects/codon_context/codon_context.py projects/codon_context/example.json` from repo root. A normal result names `constraints_pass` or `constraints_fail`, records violation positions, hashes the translated protein and returns `expression_prediction: null` with an explicit warning. A candidate that changes the amino-acid sequence raises an error rather than returning a comparable CAI gain. Missing codon weights, zero or negative weights, nonfinite values, impossible window sizes, contradictory GC bounds and ambiguous nucleotides fail closed.

The tests cover valid synonymous comparison, altered protein rejection, invalid start/stop/ambiguity, forbidden motifs, GC windows, malformed numerical constraints and all three real accession CDS checks. They test software behavior, not whether any gene expresses in a lab. No code enumerates or searches synonymous designs. This is an auditor of proposed CDS sequences, not a codon optimizer that can find a satisfying sequence on its own.

5. Evidentiary gate and missing cohort

The software gate is concrete: preserve translation, enforce explicit constraints, and never claim yield from unmeasured codon arithmetic. That gate passes the supplied tests. The original biological hypothesis remains unassessable: the later mRFP cohort provides sequence and expression for one protein/host but not the multi-host, protein-held-out, GC- and secretion-controlled design needed to test CAI-gain generalization. A stronger cohort would include original construct sequences, host strain, growth conditions, measured expression or yield, assay variability and secretion status or compatible constraints for each construct. A host-held-out and a secretion-held-out comparison would be needed to test whether CAI gain adds predictive value beyond GC and simple sequence features. A single best-performing recombinant protein or an accession-only CDS database would not suffice.

Even with a paired cohort, a robust test would freeze the unit of independence: constructs from the same protein or publication may share protocol and label noise. The final model should be assessed on a genuinely new host/lab or held-out protein family, with confidence intervals and a baseline that uses composition alone. That is future work, not a benchmark result concealed in this report. No external wet-lab measurements were acquired or inferred from UniProt/RefSeq identifiers.

6. Safety and misuse boundaries

A scientifically honest CDS checker should be predictable when input is malformed. It rejects non-ACGT bases, internal stops and candidate/reference protein mismatch before reporting a passing constraint status. It does not silently accept Boolean numeric fields or cast fractional window lengths to integers. DNA motif and GC restrictions can matter for cloning or synthesis, but the user remains responsible for choosing biologically appropriate constraints, host weights and gene identity. A passing output means only that the supplied sequence meets the supplied arithmetic rules; it is not a safety, expression, manufacturing or clinical clearance.

The tool currently assumes the standard genetic code and an ATG start. Mitochondrial genomes, organisms with alternate codon tables, noncanonical starts and engineered code systems require a separate specified mode. Secretion is not modeled, nor are mRNA secondary structure, RNA stability, rare-codon stalls, immunogenicity or regulatory sequence effects. No wet-lab phenotype can be reconstructed from the source accession metadata.

7. Reproducibility and counts

Run `python3 projects/codon_context/accession_audit.py` to refetch source records and verify translation; compare hashes in `accessions.json` before interpreting a changed UniProt or RefSeq record as the same source. Run `python3 -m pytest -q projects/codon_context` for the local software gate. The initial sequence audit used two scientific services (UniProtKB and NCBI RefSeq) and three protein-CDS pairs. Later sections add the expanded pair audit

and a distinct PSI:BiologY expression-success cohort, bringing materially used sources to three before the paired-construct scope check below. The later mRFP cohort supplies same-protein synonymous constructs and fluorescence-linked expression in one host, while the earlier expanded accession and PSI cohorts do not. It is neither a multi-host matched-yield cohort nor a matched benchmark. The cross-host CAI-gain verdict remains **unassessable**, not negative expression performance or a successful CAI predictor. A short source-backed tool report is more truthful than padding this result into an apparent 20-page expression paper.

Source notes

The [UniProtKB REST service](#) and [NCBI Entrez E-utilities](#) are the direct record sources. Exact protein and nucleotide URLs, feature coordinates and response digests are in `projects/codon_context/accessions.json`. The committed CLI tests and code establish only the claimed arithmetic checks. No accession listed here includes a paired construct/ expression value in this analysis.

Post-report CDS feature expansion, September 25

The original three pairs above remain an archival cross-species demonstration. A frozen new selection chose the first 120 reviewed E. coli K12 entries in UniProtKB search order with one unambiguous NP_ RefSeq link to NC_000913.3, before comparing translations. All 120 UniProtKB entries were fetched separately with response SHA-256; their 120 distinct CDS features were extracted from **one** fetched RefSeq chromosome (one shared response SHA-256), and each annotated protein translation matched the linked UniProt sequence. `data/expanded_selection.json` and `data/expanded_pairs.json` store exact IDs, source URLs, coordinates, sequence hashes and check results. Seven entries have valid annotated bacterial GTG/TTG starts and one has a selenocysteine translation exception. The strict ATG/standard-code constraint CLI is unchanged and must not be used to reject those valid annotated exceptions. This is 120 matched protein/CDS feature pairs, **not** 120 individually downloaded nucleotide records, independent constructs or measured expression outcomes. The no-yield-prediction gate remains in force.

Exploratory measured-expression side experiment, September 25

The original CAI-gain question remains unanswerable without matched synonymous constructs, but a secondary, distinct prediction question now uses real expression-success labels. The [Bhandari et al. 2021 article](#) links its [research source repository](#), where a fixed revision supplies the PSI:BiologY pET21_NESG sequences, species names and class table. Exact original-file URLs and digests are in `data/psi_sources.json`. The class table has 11,437 accessions, original FASTA 11,469; seven class accessions lack a supplied species name, leaving 11,430 matched records. The source authors' notebook collapses classes 1/2 into detectable expression versus 0 not found. These binary outcomes are measured expression success, not protein yield, and no two records are asserted to be synonymous versions of the same protein.

`data/expression_protocol.json` records a locally frozen design before metrics: four composition/length features, one seed-27 25%-of-prefix-groups holdout and a training-only logistic pipeline with $C=0.1$. On 8,941 train and 2,489 test records, AUROC is 0.6585; length-only 0.5365, GC-only 0.5755, intercept-only 0.5. Average precision is 0.8225 versus test prevalence 0.7453; balanced accuracy at 0.5 is 0.5123. Prefix groups are not a verified taxonomic or homology split: 39 of 130 prefixes mix named species, although the frozen split has no exact supplied species name or identical full CDS sequence on both sides. This does not establish homology independence. Source labels and experimental protocols may still be correlated, and the feature choices were motivated by prior literature. The paper's published opening-energy AUC 0.70 used its own region, methodology and cohort; it is **not a matched head-to-head comparison**. This exploratory result cannot be called a win or match. Most importantly, classification across different proteins in one E. coli expression setting says nothing about the incremental yield effect of changing synonymous codons while holding protein, host, GC and secretion constant. The CLI keeps `expression_prediction: null`.

Paired-construct scope check without model fitting

[Angov et al. 2008](#) reports wild-type versus codon-harmonized E. coli expression for five selected constructs. We read the [original supplementary Table 1](#) as an XLS with response SHA-256 in `data/synonymous_pair_scope.json`. FMP003 has one synonymous codon change and a reported harmonized purified yield of 0.07 mg/g cell paste versus wild-type not detected. The 3D7 MSP1 construct has reported wild-type 0.8 versus harmonized 3 mg/g cell paste with 246 changes. Those are selected paired intervention observations, not proof of an average CAI effect: the table lacks full paired CDS sequences and CAI, other entries include ND or Not Detected rather than quantitative zero, and there are only five reported designs from one setting. We neither fit a model nor infer a host-held-out yield response. This fourth materially used source strengthens the reason to collect construct-level paired data, while leaving the program gate open.

mRFP same-protein measured-expression scope check

The [original whole-gene codon-randomization study](#) reports 1,459 mRFP construct sequences with protein-production measurements in *E. coli*. A [downstream CodonBERT CSV](#) supplies exact 678-nt mRNA sequence, numerical expression and CodonBERT-provided train/validation/test columns. We fetched the pinned file and independently retrieved the original article supplementary workbook from [Europe PMC](#). All 1,459 original validated sequences and log-corrected fluorescence values match the downstream CSV as a multiset (maximum numeric difference 3.6e-15); the four duplicate sequences are preserved. The outer Europe PMC ZIP is regenerated, but its nested archive and workbook hashes are stable in `data/mrfp_original_crosscheck.json`. We checked the pinned CSV SHA-256 and confirmed all rows translate to the same 225-aa protein with a common stop. It contains 1,455 unique sequences: two train/test exact-sequence overlaps, one train/train duplicate and one validation/validation duplicate. We removed the two training rows whose sequences recur in the test set, averaged the remaining train/train duplicate from training measurements only, and left the 219 unique test rows untouched. This leaves 1,018 training sequences. Validation labels and values were not used for tuning. The exact-provenance and split audit are in `data/mrfp_protocol.json`, `mrfp_side_study.py` and `data/mrfp_result.json`; the raw CSV is not redistributed pending license review.

On the frozen secondary-CSV test partition, training-only standardized ridge with fixed alpha 100 has Pearson r 0.5214, MAE 0.4751 and R^2 0.2618 using whole-CDS and first-30-nt GC. A fixed positional-agnostic count of the first eight codons gives r 0.7224, MAE 0.3940 and R^2 0.5158. A training-mean constant has MAE 0.5762 and R^2 -0.0007. The response is the secondary CSV's numerical fluorescence-linked expression measurement, not purified yield. These are not matched published-model results: the original paper reports a random forest first-eight-codon correlation from its own cross-validation, with different features, splits and estimator. The direction that early codon context matters is *already known from that paper*, so this reanalysis is a leakage-aware reproduction and a useful feasibility check, **not a new biological discovery**. It supplies real same-protein construct measurements but only one protein, source laboratory and host setting; it cannot test host-, secretion- or protein-held-out generalization, or whether a change in CAI within paired designs causes improved yield. The original named-protein examples P00722, P07102 and P61769 still have zero measured synonymous constructs in this build. A post-result audit mapped each downstream row back to the original validated workbook by both DNA sequence and log response, then checked the original design library and source header as categorical predictors on the same split. Library only has test Pearson r 0.3552; original four-level header only r 0.4493. These do not explain all the first-eight-codon score (r 0.7224), but demonstrate laboratory/design structure is predictive and could leak through a within-source random split. The follow-up was designed after seeing the first result and is a sensitivity diagnostic, not an independent lab holdout or new discovery. See `data/mrfp_batch_protocol.json` and `data/mrfp_batch_result.json`. A separate **post-result** four-fold leave-original-header-out sensitivity test uses the same fixed ridge and GC/early-codon features with one of the four source headers held out in each fold. The early-codon model has test Pearson r 0.666, 0.146, 0.805 and 0.667 across the four headers, but R^2 is -0.179, -1.763, 0.641 and 0.080 respectively (unweighted macro R^2 -0.305); macro MAE is 0.480 versus training-mean 0.592. A high correlation can thus coexist with poor calibration on a held-out design batch. The four headers are *not* four independent labs, and all came from one study; this does not establish host transfer. The design and all fold outcomes are in `data/mrfp_header_holdout_protocol.json` and `data/mrfp_header_holdout_result.json`. The candidate's present contribution remains a fail-closed constraint checker and evidence/split audit, not a validated new codon-design pipeline.

Original multi-gene source test, September 26

The [2010 *E. coli* expression study](#) and its [original Europe PMC supplementary bundle](#) provide full wild-type and optimized CDS for 100 distinct protein constructs. We matched each original Table I ID, including the paper's multi-accession formatting, to its original workbook row and confirmed identical translated polypeptides up to stop for each of the 100 pairs. Optimized records append two terminal stops, while 98 WT records have one and two WT records lack a terminal stop; full raw translated strings are thus not identical. The original model retains raw-CDS GC including these short stop tails; first-30-nt GC is unchanged and the exploratory pooled R^2 remains 0.061. The table reports 73 strictly positive total-expression ratios (optimized divided by wild-type), 21 entries marked 0 (no expression) and six / entries (no analysis possible, lacking wild-type construct). The 0 and / entries are preserved in the audit but excluded from logarithmic regression, not assigned invented numeric yield. Table III contains 25 proteins with purified-yield comparisons, including multiple purification modes and undetermined wild-type values; its endpoint cannot be substituted for Table I total expression. All source hashes, the fixed analytic choices and 100 provenance-linked rows are in `data/multigene/`, with a rerunnable parser/model in `multigene_audit.py`. Original XML and XLS are fetched locally, not redistributed.

Before computing the model, we fixed three simple features (whole-CDS and first-30-nucleotide GC changes plus wild-type log length), a scaled ridge regression with alpha 10, and leave-one-original-protein-family-out testing. On the 73 positive-ratio observations in six families, pooled held-out log-ratio R^2 is 0.061 (training-mean baseline -0.003), with MAE 0.729 (mean baseline 0.753). Family-specific R^2 is negative for the 15 cytokines (-0.048); no score is a measured purified-yield score or a matched literature-model comparison. This source fills a real same-protein, multi-gene feasibility gap, but selection on measurable expression, one lab, one host, one optimizer and only six coarse families prevent a novel

generalizable CAI-gain claim. Our model has no CAI feature, cannot estimate causal effects, and does not run in the CLI's `expression_prediction` field. The fail-closed original conclusion stands; a small pooled lift over a mean baseline is not an ISEF-winning discovery.

Distinct yeast-host source, but no matched transfer endpoint

[Lanza et al. 2014](#) generated nine eGFP and ten CatA wild-type or re-coded constructs in yeast. Its [original supplementary PDF](#) lists sequences. The paper plots eGFP flow-cytometry response and CatA Vmax at three growth times and muconic-acid production; the authors report a condition-specific CatA result of 2.6-fold wild type and nearly 2.9-fold a commercial comparator in stationary-phase product yield. These are the original authors' findings, already in prior work, and unlike the 2010 E. coli paper's total-expression ratio endpoint. We did not extract numeric values from plotted figures or join different proteins as if they were the same cross-host construct. Text extraction of the supplement spans PDF page boundaries and yields inconsistent eGFP lengths, so full eGFP translation identity remains unverified here. This second host is useful for novelty and design context, **not** evidence of an independently validated host-transfer model; see `data/yeast_independent_scope.json`.

Purified-yield eligibility is smaller than total-expression eligibility

Table III of the same [original multi-gene article](#) reports 27 native/denatured purification rows for 24 unique named proteins. The prose also says 24 candidates, though the table title says 25 genes. Its original numerical rows yield 17 pairs with two strictly positive mg/L values and finite reported optimized/wild-type ratios. Six rows have `n.d.` wild-type purification and four rows have a measured zero wild-type yield, where a finite fold-change is undefined. IFN alpha, TFIIB and TNF alpha are each listed in two purification modes. `multigene_yield_scope.py` parses and checks the original ratio against the two yields; `data/multigene/yield_rows.json` preserves all 27 rows and `yield_scope.json` records counts. This selected and partly duplicated subset is too small for an honest separate protein-family or host-held-out purified-yield model. It must not be substituted for the 73 positive **total-expression** ratios in the exploratory model above.

A post-result stop-codon sensitivity test removes all terminal stops from the whole-CDS GC feature of the same original 100 pairs, without changing labels, family splits or model settings. Pooled out-of-family R2 shifts from 0.06063 to 0.06027 (difference 0.00036); first-30-nt GC remains identical. The original score's short stop-tail composition effect is small for this narrow diagnostic, but this is neither a new host benchmark nor proof that CAI controls yield. Code and result are in `multigene_stop_sensitivity.py` and `data/multigene/stop_sensitivity.json`.